

Timeouts retries and cancellation

Project: Grounded Learning Service

1. Prepare the purpose, prerequisites and materials

Teach the learning cycle

Primary target: Explain the mechanism and terminology of timeouts retries and cancellation

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Approval gates and idempotency

Without opening the old lesson, explain one key idea from Approval gates and idempotency. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of approval gates and idempotency

Reference example, not the only acceptable wording: Teacher explanation: Approval gates bind side effects to an actor, exact version and destination. Idempotency prevents repeated requests with the same key from repeating a committed effect. The mock submission requires approval for request version v1 and destination D. Store key -> committed result. Reusing the same key with the same payload returns that result; the same key with different payload is rejected.

State machines for bounded agents

Without opening the old lesson, explain one key idea from State machines for bounded agents. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of state machines for bounded agents

Reference example, not the only acceptable wording: Teacher explanation: A bounded agent uses explicit states and finite action limits. A state machine makes invalid transitions visible. States are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised evidence. done is terminal. Maximum two lookup attempts.

Evaluation pipelines

Without opening the old lesson, explain one key idea from Evaluation pipelines. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of evaluation pipelines

Reference example, not the only acceptable wording: Teacher explanation: Evaluation pipelines preserve case versions, procedure versions and raw outcomes before producing metrics. A rerun must identify what changed. Load versioned cases, validate expected outcomes, execute the frozen system, store case-level results and compute declared denominators. Reject missing results rather than silently counting them as success.

Two 45-minute sessions. Timeouts bound waiting; retries consume a finite budget. Cancellation should stop future work while accounting for operations already committed. Use the supplied constructed scenarios to make the mechanism inspectable. Collect W4 before sharing solutions. Students should name intermediate evidence and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the mechanism and terminology of timeouts retries and cancellation

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

Prerequisite answer

A result depends on the inputs, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from predictions or unmeasured conditions. If the prior concept cannot be explained, use a short retrieval activity before this chapter.

End goal for the pictorial case

Explain timeouts retries and cancellation using the deciding step in this fictional service.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain and model the mechanism

Timeouts bound waiting; retries consume a finite budget. Cancellation should stop future work while accounting for operations already committed.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Retry only declared transient failures. Use at most three attempts with waiting gaps 1 and 2 seconds; each attempt may run at most 4 seconds. Worst-case elapsed bound excluding overhead is $3 \cdot 4 + 1 + 2 = 15$ seconds.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term · Meaning

Term: Timeout

Meaning: A declared limit on waiting

Term: Retry budget

Meaning: The maximum repeat attempts or resources

Term: Cancellation

Meaning: A request to stop future processing

Timeouts retries and cancellation — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Three attempts each time out after 4 seconds with gaps 1 and 2. A fourth is proposed.

Check or calculation: Retry only declared transient failures. Use at most three attempts with waiting gaps 1 and 2 seconds; each attempt may run at most 4 seconds. Worst-case elapsed bound excluding overhead is $3 \cdot 4 + 1 + 2 = 15$ seconds.

Conclusion and limit: Budget is exhausted after 15 seconds under the model; reject fourth attempt. A late side-effect result must still be reconciled.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Budget is exhausted after 15 seconds under the model; reject fourth attempt. A late side-effect result must still be reconciled. Qualification: A timeout means the caller lacks a timely result, not necessarily that the remote operation did not occur. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Three attempts each time out after 4 seconds with gaps 1 and 2. A fourth is proposed.

Budget is exhausted after 15 seconds under the model; reject fourth attempt. A late side-effect result must still be reconciled.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary input, record the outcome and state the contract or statistical assumptions. The program

uses fictional values and the Python standard library; it performs no external operations.

```
def waiting_bound(attempts, per_attempt, gaps):  
    if type(attempts) is not int or attempts < 1 or len(gaps) != attempts -  
1:  
        raise ValueError('attempt budget')  
    if per_attempt < 0 or any((g < 0 for g in gaps)):  
        raise ValueError('negative duration')  
    return attempts * per_attempt + sum(gaps)  
  
print(waiting_bound(3, 4, [1, 2]))
```

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Timeouts retries and cancellation

Retry within declared limits. A read-only lookup allows two attempts of at most one second each. Cancellation forbids further attempts.

The facts we will use

First attempt times out at 1 second. Second succeeds in 0.4 seconds. Ignore overhead. No cancellation occurs.

Timeouts retries and cancellation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Bound first wait

Stop attempt one after its one-second allowance.



2. Check retry

One attempt remains, and cancellation has not occurred.



3. Complete next attempt

Accept the checked record; total waiting is 1.4 seconds.

CHANGE ONE THING

Cancellation arrives after the first timeout.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Bound first wait

What happens and why: Stop attempt one after its one-second allowance.

Step: 2. Check retry

What happens and why: One attempt remains, and cancellation has not occurred.

Step: 3. Complete next attempt

What happens and why: Accept the checked record; total waiting is 1.4 seconds.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A timeout bounds waiting; a retry tries again; cancellation withdraws the request. They are separate conditions on the next action.

What this example does not establish

The arithmetic excludes overhead; retries of side-effecting operations require separate duplicate and recovery controls.

Change one thing in the pictured case

Change: Cancellation arrives after the first timeout.

Prediction: Do not start the second attempt.

Reason: Cancellation overrides the unused retry allowance.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Time or stage · Teacher action

Time or stage: Session 1: 0–7 minutes

Teacher action: Retrieve the prerequisite; identify the input, procedure and evidence source.

Time or stage: 7–18 minutes

Teacher action: Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.

Time or stage: 18–32 minutes

Teacher action: Trace the declared procedure and worked case. Complete W2, recording intermediate

steps.

Time or stage: 32–45 minutes

Teacher action: Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

Time or stage: Session 2: 0–8 minutes

Teacher action: Retrieve the worked reasoning and collect the guided response before discussing it.

Time or stage: 8–20 minutes

Teacher action: Compare a proposed boundary variant with the original case; identify what remains fixed.

Time or stage: 20–35 minutes

Teacher action: Collect W4 independently. Require the procedure trace and the bounded conclusion.

Time or stage: 35–45 minutes

Teacher action: Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Reduce help deliberately

Completion task: Three attempts each time out after 4 seconds with gaps 1 and 2. A fourth is proposed. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Budget is exhausted after 15 seconds under the model; reject fourth attempt. A late side-effect result must still be reconciled.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Timeouts retries and cancellation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A timeout means the caller lacks a timely result, not necessarily that the remote operation did not occur.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Calculate waiting time for a one-second timeout followed by a 0.4-second success. What changes if cancellation arrives after the first attempt but before the retry?

Reasoned teacher answer

Stop attempt one after its one-second allowance. One attempt remains, and cancellation has not occurred. Accept the checked record; total waiting is 1.4 seconds. Changed case: Do not start the second attempt. Cancellation overrides the unused retry allowance.

Changed-case reasoning: Do not start the second attempt. Cancellation overrides the unused retry allowance.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the investigation and diagnose misconceptions

The worked result establishes every future case: A timeout means the caller lacks a timely result, not necessarily that the remote operation did not occur.

A missing input may be replaced without recording the change: A replacement changes the evidence or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent evidence: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Guided and exit questions

Guided: Budget is exhausted after 15 seconds under the model; reject fourth attempt. A late side-effect result must still be reconciled. Qualification: A timeout means the caller lacks a timely result, not necessarily that the remote operation did not occur. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Teacher explanation: Timeouts bound waiting; retries consume a finite budget. Cancellation should stop

future work while accounting for operations already committed. Retry only declared transient failures. Use at most three attempts with waiting gaps 1 and 2 seconds; each attempt may run at most 4 seconds. Worst-case elapsed bound excluding overhead is $3 \times 4 + 1 + 2 = 15$ seconds.

W2 Trace the worked case

Three attempts each time out after 4 seconds with gaps 1 and 2. A fourth is proposed. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Budget is exhausted after 15 seconds under the model; reject fourth attempt. A late side-effect result must still be reconciled.

W3 Audit the limit

A report omits this qualification: A timeout means the caller lacks a timely result, not necessarily that the remote operation did not occur. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A timeout means the caller lacks a timely result, not necessarily that the remote operation did not occur. State the observed result separately from the unverified or out-of-scope claim; preserve the original evidence rather than inventing a missing fact.

W4 Apply a new case independently

Two attempts, each up to 5 seconds, gap 2. Compute bound and decide whether authentication failure should retry.

Teacher explanation: Bound 12 seconds excluding overhead. Authentication failure is non-transient under this policy; do not retry it.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked input or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure. Require preserved settings, original outputs and data roles. A timeout means the caller lacks a timely result, not necessarily that the remote operation did not occur. The new case should test this particular mechanism rather than merely restate the original result.

Independent assessment and fresh retry

W4: Bound 12 seconds excluding overhead. Authentication failure is non-transient under this policy; do not retry it. Require intermediate reasoning and the stated scope. Fresh retry: Four attempts, each up to 2 seconds, gaps 1, 2, 4. Expected reasoning: Bound 15 seconds. Cancellation prevents new attempts; it cannot undo an already committed effect. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case: Two attempts, each up to 5 seconds, gap 2. Compute bound and decide whether authentication failure should retry.

Selected case answer

Teacher explanation: Bound 12 seconds excluding overhead. Authentication failure is non-transient under this policy; do not retry it.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect the project, support and next lesson

Add a labelled timeouts retries and cancellation evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of timeouts retries and cancellation. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: External side effects and recovery. Preserve the current case record and retrieve its evidence boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-6/17, ai-6/19, ai-6/22. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.